

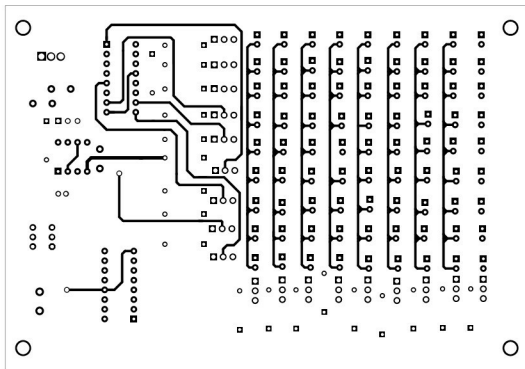


Fig. 4: Actual-size, top-side of the PCB shown in Fig. 3

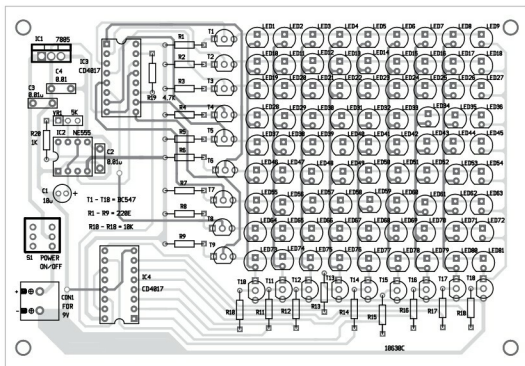


Fig. 5: Component side of the PCB shown in Fig. 3

IC2 generates clock for the first decade counter (IC3). This counter controls the rows (anodes of LEDs) with the help of nine BC547 transistors (T1 through T9). These transistors can be controlled by output pins (O0 through O9) of IC3 along with 220-ohm resistors (R1 through

R9) that are connected to collectors of T1 through T9 for current-limiting purpose.

Tenth output (O9) is fed to pin 14 (CLK) of CD4017 (IC4) as clock input. Columns of the LEDs can be controlled through collectors of BC547 transistors (T10 through T18) that are connected to cathodes of LED columns. Outputs of IC4 are connected to bases of transistors T10 through T18 through resistors R10 through R18, respectively.

Clock rate can be varied with the help of 5-kilo-ohm potentiometer (VR1) to produce stunning running lights effect based on the concept of persistence of vision.

Construction and testing

Actual-size, double-side PCB layouts for the 81-LED chaser light using two CD4017 ICs are shown in Figs 3 and 4, respectively, and their component layouts in Fig. 5. After assembling the circuit on the PCB, enclose it in a suitable plastic box.

The circuit works off a 9V battery. **EFY**



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